

EffectSlice: Evidence-Carrying Admission for Paper-Derived Procedural Cores

Anonymous submission

Abstract

Turning a paper into a compact agent artifact does not make that artifact a reliable substitute: source provenance and behavioral sufficiency are different claims. We introduce **EffectSlice**, a reducer-agnostic candidate-admission protocol for paper-derived procedural cores. Given a no-artifact baseline (B), registered full artifact (F), and selected strict subset (S), it binds instructions to paper spans and dependencies, freezes the candidate, permits visible public testing, and scores each fresh terminal run once with a private suite. Admission is a hash-bound finite-schedule decision, not a population estimate. Across two sequentially registered schedules and 234 fresh DeepSeek endpoint conversations, the protocol separates three decision states: SNAP prefix 03 has insufficient F and S (0/15/5 B/F/S successes, Reject); Toolformer prefix 01 has sufficient F but insufficient S (0/17/8, Reject); in a separately registered later schedule, a local admission is issued for pre-existing Toolformer prefix 04 (0/18/17, Pass) on a new nonoverlapping private registry. A planted redundancy smoke test and byte-identical identity instrument pass their task-local checks. The resulting one-of-three admission is a descriptive count across sequential schedules, not a prospectively registered success rate, and does not establish broad automatic reduction, paper-level equivalence, or human benefit.

Introduction

Paper-reproduction agents increasingly turn research descriptions into code and executed experiments (Starace et al. 2025; Zhao et al. 2025), while paper-to-agent systems package papers and associated software as interactive tools (Miao et al. 2025). In parallel, agent-skill systems compile external resources into reusable instructions (Shen et al. 2026; Pan et al. 2026). A practical workflow is now plausible on an ordinary PC: call a vendor model API, run code locally, and use a paper-specific procedure without local model deployment.

The difficult step is deciding what the reusable *core* is. A paper may describe a five-step mechanism, while a reducer retains only the first three steps because they were sufficient on one development example. The resulting artifact can be

short and traceable to the paper without being a reliable substitute for the complete procedure. Conversely, a complete artifact can be correct but inconsistently used by a bounded agent. These are behavioral questions that token counts and source overlap do not answer.

Recent work makes this tension concrete. SkillReducer reports that large-scale skill compression can improve functional quality (Gao et al. 2026). A controlled 1,200-rollout decomposition instead finds no representation strategy that improves over raw skills and identifies executor capability as the dominant factor (Xu and Wu 2026). Both are important optimization studies. Our question is downstream and candidate-specific: after any system proposes a paper-derived core, what evidence is required before a user is told that it can stand in for the complete paper-derived artifact?

Three requirements follow. First, *source-addressed provenance* must state exactly which paper spans support each executable instruction and which prerequisites were retained. It does not certify a complete or correct interpretation. Second, evaluation must preserve a usable public feedback loop while preventing the private score from becoming another development tool. Third, the decision rule itself needs controls: a planted redundant unit should be removable, a preregistered structurally incomplete candidate expected to fail should be rejected, and byte-identical artifacts should reveal measurement noise rather than being assumed equivalent across fresh agent sessions.

We introduce **EffectSlice**, an evidence-carrying admission protocol rather than a new compression algorithm. It represents a paper-derived procedure as source-addressed atoms, freezes a dependency-closed strict subset, and runs three conditions: no artifact (B), the full artifact (F), and the selected slice (S). Each condition receives a fresh provider conversation. The public test is visible and callable; the private 64-case scorer is invoked only on the fixed terminal patch and its result is not returned for another model turn. Separate pre-run and post-run manifests bind the task, paper source, atoms, artifacts, tests, scorer, runner, transport, schedule, outputs, and analysis lineage by digest.

The two schedules jointly answer three decision questions: whether admission is possible when F itself is insufficient (RQ1), whether an S should be rejected despite sufficient F (RQ2), and whether a nearer pre-existing S can earn a local admission on a separately registered later holdout (RQ3).

This is an anonymized submission for review purposes only. Distribution, citation, or public sharing of this manuscript is strictly prohibited. Copyright and publication details will appear in the final version if accepted.

Our contributions are:

- We formalize source-addressed procedural atoms with exact source locations, executable roles, dependency edges, and immutable artifact identities, separating provenance from behavioral sufficiency.
- Our main contribution is the candidate-level admission record: candidate and source identities, public/private information boundary, frozen schedule, immutable outcomes, and a binary finite-schedule Admit/Reject decision. Reject means only “do not admit under this registered boundary,” and makes no IID, cross-paper, or population claim.
- We execute two sequentially hash-registered schedules with three non-planted candidates from our project-local deterministic prefix reducer, a planted smoke test, a registered negative control, and identity instrumentation. V4 admits neither candidate: SNAP’s registered full artifact itself misses sufficiency, while Toolformer prefix 01’s slice misses sufficiency despite a sufficient full artifact. A later schedule admits a pre-existing four-of-five-atom candidate. We preserve all 234 conversations, terminal patches, response identifiers, test traces, scores, and digests, and disclose and outcome/count cross-check two post-run analyzer compatibility repairs.

We do not claim a general compression advantage, lower cost, local-model inference, full-paper reproduction, or benefit to human users.

Related Work

Paper reproduction and executable research. PaperBench evaluates agents that recreate AI research repositories (Starace et al. 2025); AutoReproduce adds paper lineage to automated experiment reproduction (Zhao et al. 2025); and Paper2Agent builds interactive agents around papers and code (Miao et al. 2025). These systems primarily ask whether a complete implementation or interface can be constructed. EffectSlice starts after an author-constructed registered full artifact exists and audits a proposed strict subset.

Skill construction, retrieval, and compression. SkillFoundry and Anything2Skill compile reusable skills from heterogeneous sources (Shen et al. 2026; Pan et al. 2026). Graph-of-Skills uses dependency-aware retrieval (Liu et al. 2026), SkillRAE compiles compact execution context (Meng, Wang, and Fang 2026), and ClawTrace attributes cost within skill-distillation traces (Yuan et al. 2026). SkillReducer compresses routing descriptions and skill bodies, then restores content when task feedback shows a regression; it also reports an independent deterministic benchmark (Gao et al. 2026). Xu and Wu freeze ten skill-delivery conditions over 40 software tasks, compare raw, shortened, structured, and scoped forms, and use task-clustered inference and real-cost accounting (Xu and Wu 2026). Shaposhnikov et al. provide our strongest utility baseline by comparing behavior with and without whole skills (Shaposhnikov et al. 2026); they do not decide whether a source-bound strict subset can substitute for a registered

full artifact. EffectSlice neither competes with their reducers nor repeats their cost question. It adds source-span binding and a candidate-level admission record after the candidate is frozen.

Reduction and adaptive evaluation. Delta debugging and C-Reduce remove input components while preserving an observed behavior (Zeller and Hildebrandt 2002; Regehr et al. 2012); cause reduction extends the idea beyond crashes (Groce et al. 2016). Prompt compression similarly trades context size against retained information (Jiang et al. 2023; Pan et al. 2024). Repeated holdout exposure can, however, overfit an adaptively selected hypothesis (Dwork et al. 2015; Nakkiran and Błasiok 2018). EffectSlice applies this information-flow lesson to tool-using code agents: public tests support development, while the fixed patch receives one terminal private evaluation. The increment is binding that isolation to paper-source addresses, B/F/S candidate identities, and an archival local decision record.

Problem Setting

Let paper P have a cryptographic identity d_P . We represent a bounded paper-specific procedure as atoms $A = \{a_1, \dots, a_K\}$ and a directed acyclic dependency graph G . Each atom stores

$$a_i = (\text{id}_i, \ell_i, s_i, r_i, D_i), \quad (1)$$

where ℓ_i is an exact source locator (document digest, byte and line spans), s_i is the executable instruction, r_i its contract role, and D_i the set of atom IDs of its direct dependencies. A candidate atom set $A_S \subsetneq A$ is nonempty and dependency closed iff $D_i \subseteq \{\text{id}_j : a_j \in A_S\}$ for every $a_i \in A_S$. A deterministic renderer R , using topological order with atom-ID tie breaking, produces the full artifact $F = R(A)$ and candidate artifact $S = R(A_S)$. Both are registered before confirmation. This representation records provenance; it does not assert semantic completeness or that an agent will use the instructions.

For registered block r , condition $C \in \{B, F, S\}$ starts a fresh conversation with the same task, an isolated copy of the same starter workspace, action interface, public test, action budget, and private scorer. Only the artifact context differs. Let $q_{C,r} \in [0, 1]$ be the terminal private score, $t_{C,r}$ indicate the hard task contract, and $p_{C,r}$ indicate that the terminal patch was applied. Operational success is

$$z_{C,r} = \mathbb{I}[t_{C,r} = 1 \wedge p_{C,r} = 1 \wedge q_{C,r} \geq 0.95]. \quad (2)$$

A threshold of 0.95 over 64 deterministic cases requires at least 61/64 cases. The raw scorer’s `success` flag means exact 64/64 success; every success count in our tables and decisions instead uses the operational definition above. Result summaries and Figure 1 use *Pass* as a synonym for *Admit*. An exhausted transport retry, unapplied patch, or hard-contract failure is a zero operational success and remains in the registered denominator. A missing or malformed registered output makes the schedule Invalid/No record; it cannot be used as either admission or rejection evidence.

Study	Artifact and objective	Evaluation boundary	Distinguishing output
SkillReducer (Gao et al. 2026)	Generic agent skills; token-efficient routing and progressive disclosure	Self-correcting task feedback plus an independent deterministic skill benchmark	Optimized skill and aggregate retention/quality
Xu and Wu (Xu and Wu 2026)	Raw and transformed skills; quality/cost factor decomposition	40 tasks, 10 conditions, three repetitions; task-clustered contrasts	Multi-task aggregate condition comparison and cost diagnosis
Agentic Skills Framework (Shaposhnikov et al. 2026)	Whole executable skills; marginal utility	Scalable with-skill versus without-skill task evaluation	Aggregate whole-skill utility
EffectSlice (ours)	Source-addressed atoms from each source paper; admit or reject one fixed strict subset	Visible public test, one terminal private score per fresh condition run, complete registered finite schedule	Hash-bound candidate record with a task-local admission or rejection

Table 1: Closest-work comparison. EffectSlice is a downstream admission and audit layer, not a claim of superior compression or broader empirical coverage.

EffectSlice makes a finite engineering decision. For an 18-block family, let $k_C = \sum_r z_{C,r}$. The candidate is admitted exactly when

$$\begin{aligned} k_B &\leq 2, & k_F &\geq 16, & k_S &\geq 16, \\ k_F - k_S &\leq 2, \end{aligned} \quad (3)$$

and every registered output and schedule-level integrity check is complete. Equation 3 is a Git-committed, hash-anchored service-level rule for this schedule. For a complete valid schedule that misses this gate, the record says Reject: it withholds admission under this frozen boundary rather than asserting that the candidate is semantically invalid. The binary decision applies only after the completeness and integrity gate; malformed schedules produce no decision record. It is not a confidence interval, prevalence estimate, or hypothesis about a population of papers or provider sessions.

The four numeric fields use a uniform two-run tolerance chosen from prior development evidence and frozen before V4; they are deployment parameters, not universal constants or values fitted to V4 outcomes. At $n = 18$, the registered $k_F - k_S \leq 2$ field is logically implied by $k_S \geq 16$ and $k_F \leq 18$. We retain it because it was preregistered, but do not count it as independent evidence. Admission is therefore a count-level SLA, not block-level behavioral equivalence.

The evidence-carrying output is a record

$$\mathcal{E} = (d_{\text{pre}}, M_{\text{post}}, \sigma, \mathcal{O}, \delta), \quad (4)$$

where the pre-run commitment root d_{pre} recursively binds d_P , the atom map and graph, rendered artifacts, task, public and private tests, scorer, bound analyzer, runner, transport, schedule σ , and their own digests. After execution, the digest-indexed archival manifest M_{post} binds d_{pre} , immutable output set $\mathcal{O}$, raw-evidence digest, analysis and compatibility code, independent crosscheck, generated result macros, and local decision δ . The supplied `effectslic-paper-results-manifest.v5` file `generated_results_manifest_v5.json` is canonical; older numbered manifests are superseded. It preserves lineage but does not retroactively preregister post-run repairs or derived artifacts. A score without these candidate, commitment, and information-boundary identities is not an EffectSlice admission record.

EffectSlice

Source-Addressed Candidate Construction

The extraction stage creates the atom registry from quoted paper passages and records dependencies separately from prose order. A reducer may be heuristic, learned, or interactive; the admission protocol only requires a deterministic candidate identity and closure audit. SNAP prefix 03 and Toolformer prefix 01 were locked before V4 provider calls. Toolformer prefix 04 already existed in the same deterministic reducer registry, but we selected it after inspecting V4, then froze it before V5. This project-local ordered prefix reducer is neither our contribution nor an implementation of SkillReducer.

The closure auditor rejects unknown atom IDs, graph cycles, a non-strict set, or any retained atom whose transitive prerequisite is absent. It then renders with the registered topological tie break and verifies d_F and d_S . Every V4 and V5 strict-subset family passed these checks before its provider execution. The atom boundaries and dependency edges were author-engineered from the cited passages; their semantic correctness was not independently adjudicated.

Table 2 exposes the complete atom inventory. The SNAP candidate retains preprocessing and degree construction but omits the linear-operator and eigensolver/guard atoms. Toolformer prefix 01 retains only the future-token weight schedule; prefix 04 retains the loss, counterfactual, and threshold but omits the all-candidate/order unit. They are concrete core hypotheses. Prefix 04 was selected using V4 results, so its claim rests only on the subsequently registered V5 block, not on prospective V4 selection.

Public Development, Terminal Private Scoring

The model sees the task, common action protocol, assigned artifact, an isolated starter-workspace copy, and the public test file. Within a 16-action budget it may search, open, edit, and run the public test. Public output is returned and may guide another model turn. A `submit` action freezes the patch, invokes the private 64-case scorer once, stores the result, and terminates the conversation. If the action budget ends with a nonempty patch, that final patch is scored once;

Task/family	Registered source-addressed units in F	Fixed S
SNAP-MFSE	A01 IDF scaling; A02 row normalization; A03 self-similarity removal and degree; A04 matrix-free linear operator; A05 eigensolver, ordering, and no-materialization guard. <i>Source/deps: extracted lines 73–79; A01 $\prec$ A02 $\prec$ A03 $\prec$ A04 $\prec$ A05.</i>	A01–A03 (3/5)
Toolformer prefix 01	T01 normalized future-token weights; T02 weighted loss; T03 stronger counterfactual; T04 inclusive threshold; T05 all-candidate/order rule. <i>Source/deps: lines 76–79, 134–171, 268–272; T01 $\prec$ T02 $\prec$ T03 $\prec$ T04 $\prec$ T05.</i>	T01 (1/5)
Toolformer prefix 04	T01–T05 and source/dependency chain as above. This candidate was a pre-existing output of the registered deterministic prefix reducer.	T01–T04 (4/5)
Planted positive	T01–T05 above plus T06, a registered restatement of the T04/T05 invariant that adds no computation. <i>T06 cites lines 76–79 and 159–171 and requires T04/T05.</i>	T01–T05 (5/6)

Table 2: Explicit candidate boundary with inspectable extracted-text locators and dependency order. Registries additionally store byte spans, roles, and file digests. The identity family uses byte-identical F and S and is instrumentation, not a strict-subset candidate.

an empty patch fails without feedback. No private result is sent back to the model.

The family manifest binds the paper text, atom map, F , S , public test, private case registry, scorer, starter workspace, prompt, runner, analyzer, transport, thresholds, block identifiers, and B/F/S order by SHA-256. Output directories are write-once. Provider response IDs must be unique, and every transport attempt is retained. This yields a trace from a paper span to the artifact seen by the model, its tool trajectory, the fixed patch, and the final score.

Controls and Interleaved Schedule

The V4 schedule contains 18 temporal strata and at most two workers. Each stratum contains the SNAP real candidate, the Toolformer real candidate (also the registered negative control), and the planted positive family. Six distributed strata additionally contain identity instrumentation. Each 18-block family uses all six B/F/S orders exactly three times; the identity family uses each order once. In total there are 60 blocks and 180 fresh condition conversations.

After V4, we selected the nearest untested pre-existing Toolformer reducer output, prefix 04. V5 separately freezes 18 blocks over a new private registry, with all six B/F/S orders used three times and 54 fresh condition conversations. Its cases are disjoint from all earlier private cases by ID, seed, and payload digest. V5 therefore tests a post-V4 candidate on new evidence; it is not an extension of the V4 preregistration.

The planted family is a pipeline smoke test, not evidence of useful compression or naturally redundant paper text. Calibration passes only if it is admitted, the real negative is rejected, identity instrumentation passes, all outputs exist, and integrity holds. Identity passes when at least five of six F/S success indicators match and the aggregate F/S success-count gap is at most one. At six pairs the latter is implied by the former and is retained as a registered consistency field, not independent evidence. Identity does not require the two fresh sessions to return the same patch or score.

Experimental Setup

Paper-specific tasks. SNAP-MFSE isolates SnapATAC2’s matrix-free spectral embedding core (Zhang et al. 2024): IDF scaling, row normalization,

self-similarity removal, degree construction, normalized operator application, and leading spectral coordinates. Toolformer filtering isolates the rule for retaining API calls (Schick et al. 2023): future-token weights, weighted losses, the stronger counterfactual, inclusive thresholding, and ordered application to all proposals. Each V4 or V5 private block contains 64 deterministic cases and hard interface/guard checks. Cases characterize one terminal patch and are not treated as independent model runs.

Freeze and holdout chronology. V3 development evidence informed the candidates and the uniform two-run SLA tolerance. We then froze F , A_S , S , both controls, thresholds, and the 18-stratum schedule in Git commit ae97dff, before any V4 provider call. The V4 builder generated new private case blocks and rejected overlap with every earlier private block by case ID, seed, and case-payload digest. V4 private payloads did not select its candidates. Visible public tests were fixed before private execution but are development feedback, not holdout evidence.

After observing V4 outcomes, we selected Toolformer prefix 04 from the pre-existing reducer registry and preregistered V5 in commit da0ee5d. Attempt 1 stopped during local preflight because the generic runner did not yet allow the new case-block label; it made zero provider calls and wrote no block outputs. Commit 6538e8d added that explicit allowlist entry, froze a new evidence boundary and preregistration digest, and preceded every V5 provider call. The V5 builder again excludes overlap by ID, seed, and payload digest.

Provider and host boundary. Inference used the user-supplied trusted DeepSeek endpoint. We report the provider/user label “DeepSeek V3.2” and exact request alias deepseek-v4-flash; the alias-to-deployment mapping is not independently attested. Requests used OpenAI-compatible chat completions, temperature zero, an 8192-token turn limit, a 240-second timeout, and at most five transport attempts. The provider exposes no supported random seed, so “fresh” means a new top-level conversation, not independent or reproducible backend sampling.

A Windows PC with a Ryzen 9 7945HX, 31.7 GiB RAM, and an 8 GiB RTX 4060 Laptop GPU performed

orchestration, bounded execution, testing, scoring, and persistence. Model inference stayed at the vendor API; the local GPU was not used. This is an API-first ordinary-device setting, not a local-quantized-model result.

Results

Three Decision Questions

Figure 1 and Table 3 report both complete registered schedules.

RQ1: Can a candidate be admitted when F is insufficient?

In V4, SNAP B/F/S succeeds in 0/15/5 blocks. The full artifact misses the minimum by one, the slice misses it by eleven, and the F–S count gap is ten. The result therefore finds that neither F nor S meets its frozen sufficiency requirement for this schedule.

RQ2: Does the gate reject S when F is sufficient?

Toolformer B/F/S succeeds in 0/17/8 blocks. Here the full artifact clears its requirement, but the one-atom slice does not and trails F by nine successes. The mean private score is 0.944 for F and 0.481 for S. Thus F meets the frozen sufficiency requirement while B does not, but the proposed core fails its own requirement. Neither V4 non-planted candidate is admitted.

RQ3: Can a nearer pre-existing S earn local admission?

Prefix 04 existed before V4 but was selected after V4 analysis; V5 then froze and evaluated it on new private cases. It retains T01–T04 and omits T05. B/F/S succeeds in 0/18/17 blocks, with means 0.000/1.000/ 0.944. All six frozen SLA and integrity fields pass, so the non-planted strict subset is admitted for this task, provider, and private registry. The result does not imply that T05 is dispensable for Toolformer in general; it says this fixed artifact substitutes under the registered V5 boundary.

The omitted atoms offer a plausible diagnosis but not a causal ablation. SNAP’s slice stops before the matrix-free operator and solver guard; Toolformer prefix 01 stops before every loss and decision unit. V4 failures are consistent with those omissions, but no atom-restoration ablation identifies a cause. Prefix 04 omits only the all-candidate/order unit and passes V5; that is task-local sufficiency evidence, not a causal or paper-wide redundancy claim.

Controls Calibrate the Decision

The preregistered `expected_admission` fields are expectation metadata only: they encode calibration expectations for controls and planning expectations for other candidates. They never enter the admission decision, which is recomputed solely from registered outputs and SLA fields.

The planted family removes only redundant T06. It yields B/F/S counts 0/18/17, an F–S gap of one, and passes every registered field. The same scorer and Toolformer task reject the real one-atom candidate at 0/17/8. Thus the protocol does not reject every strict subset, and its registered positive and negative expectations are both observed. This is a harness smoke test, not evidence that a non-planted compact core has been found. That stronger, still task-local evidence comes

from V5 prefix 04: it was present in the reducer registry before V4, selected after V4, frozen before V5, and admitted on new private cases. This chronology avoids V5 holdout reuse while openly retaining post-V4 selection bias.

For identity instrumentation, F and S are byte-identical yet their fresh conversations succeed in 6/6 and 5/6 blocks. Five of six paired success indicators match and the aggregate gap is one, so the instrument passes. The single mismatch is important: artifact identity does not imply trajectory or patch identity under fresh provider sessions. This motivates a tolerance-based finite SLA rather than exact per-pair equality.

Together, planted-positive admission, real-negative rejection, and identity instrumentation make the overall calibration decision Pass. This is a task-local check of the measurement stack, not sensitivity or specificity over a population of skills.

Public Iteration and Evidence Cost

The two schedules contain 234 conversations, 1760 provider turns, and 1760 recorded actions. The agent requested public tests 291 times; 278 passed and 13 did not. V5 alone contains 381 actions, 17 invalid actions, one nonpassing public test, and 2 extra transport attempts. Every V5 conversation explicitly submitted; all V4 and V5 outputs are complete.

The provider reported 2,812,362 input and 2,894,613 output tokens across 1,776 transport attempts. These totals document evidence cost, not an efficiency advantage. The protocol deliberately spends repeated API calls to audit substitution rather than claiming a shorter skill is cheaper.

Analyzer and Release Integrity

The V4 and V5 bound analyzers stopped before aggregation because a copied check conflated `comparison_role=development_triage` with the formal `evidence_boundary`. Narrow post-hoc adapters validate both raw fields, normalize only that in-memory comparison, and leave preregistrations, runners, transcripts, patches, scores, thresholds, schedules, and raw manifests unchanged. Independent outcome/count crosschecks import neither analyzer nor adapter: they reread 660 V4 and 198 V5 raw files, verify terminal private-score and response-ID invariants, and reproduce every reported count, mean, resource total, control, and decision. The canonical post-run manifest records its pre-run roots, both raw inventories, analysis lineage, tests, and generated results. A separate release verifier rehashes all 858 raw files plus 24 directly bound artifacts. These checks do not rerun the scorer or independently validate the semantic task contract.

Discussion

Source-addressed provenance is needed but is not sufficiency. A source span can prove that an instruction came from the paper. It cannot prove that a selected set contains every operation an agent needs or that the agent will execute them reliably. EffectSlice connects provenance to a behavioral record without conflating them: source atoms answer “why is this instruction here?”, while the B/F/S schedule answers “may this fixed subset substitute for the complete artifact here?”

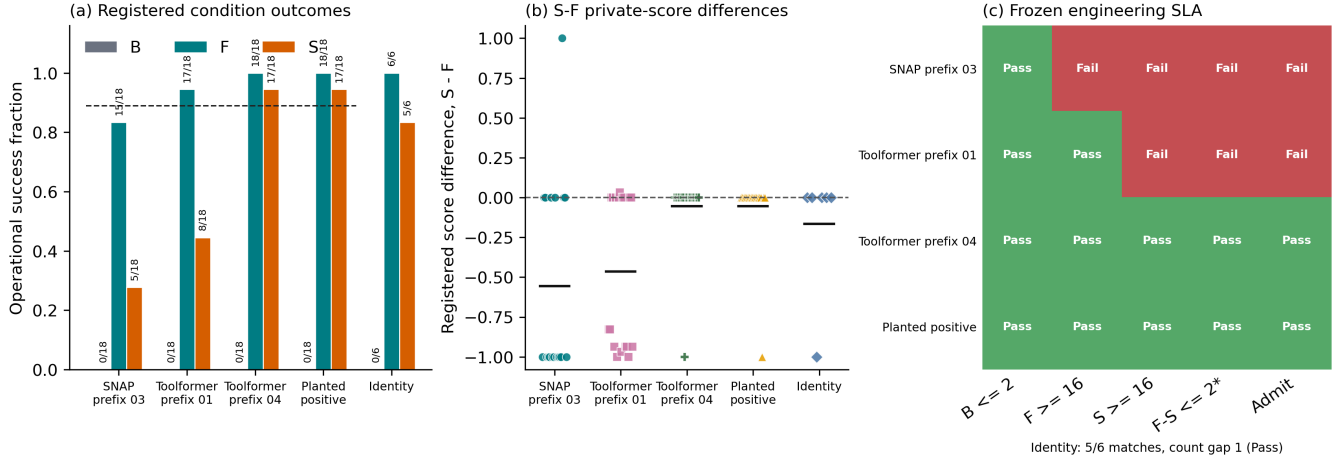


Figure 1: Complete V4 evidence plus the separately registered, later V5 candidate. (a) B/F/S operational successes; labels show exact counts, and the dashed line marks the 16/18 requirement for the four 18-block strict-subset families. (b) Per-block $S - F$ private-score differences; black lines are family means, not population estimates. (c) Frozen SLA fields and decisions. The starred field is implied by $S \geq 16$ at $n = 18$; identity uses a separate rule.

Family	Blocks	B success	F success (mean q)	S success (mean q)	F-S gap	Decision/status
SNAP real	18	0	15 (0.877)	5 (0.321)	10	Reject
Toolformer real/negative	18	0	17 (0.944)	8 (0.481)	9	Reject
Toolformer prefix 04	18	0	18 (1.000)	17 (0.944)	1	Pass
Planted positive	18	0	18 (1.000)	17 (0.944)	1	Pass
Identity instrument	6	0	6 (1.000)	5 (0.833)	1	Instr. Pass

Table 3: Registered outcomes across sequential V4 and V5 schedules. Admission uses the frozen count-level SLA plus complete output and integrity. The F-S field is retained but redundant at these thresholds. Identity is not an admission and uses its separate 5/6 match rule.

The contribution is admission, not compression. SkillReducer’s self-correcting optimizer can restore content after task failures (Gao et al. 2026); Xu and Wu show that representation changes can be dominated by executor capability (Xu and Wu 2026). Either pipeline could feed a candidate into EffectSlice. Our result does not overturn their aggregate findings. It shows that one compact candidate should carry a source identity, an information-flow boundary, and a local admit/reject record before reuse.

A local admission or rejection is the reusable output. For a nonexpert workflow, the useful output need not always be a smaller skill. If F is not sufficiently reliable, retain the full artifact with a warning. If F works but S fails, expose which source units were omitted and do not advertise S as the paper’s core. If S passes, publish it together with its task, provider, scorer, schedule, and evidence boundary. This defines an auditable output contract; whether it helps people complete real projects faster remains an open user-study question.

Limitations and Broader Impact

The study covers two papers, two bounded implementation tasks, one provider endpoint, one action scaffold, and two sequential registered schedules. The tasks exercise paper-

specific mechanisms rather than complete SnapATAC2 analyses or Toolformer training. The simple prefix candidates are intentionally weak reducers; we neither run nor claim an official external reducer implementation. The positive control removes planted redundancy. Broader candidate generators, papers, tasks, providers, and independently designed registries are required before any effectiveness claim.

The atom boundaries, executable instructions, dependency graph, and the claimed completeness of F were not independently reviewed. Their digests prove identity, not semantic correctness. The two V4 candidates have structurally obvious omissions. V5 admits one closer, non-planted Toolformer subset, but it was selected after V4 and tested on only one new registry for the same task. This is not cross-paper or paper-wide equivalence. Finally, the count-level SLA does not establish block-level trajectory preservation.

Fresh conversations are not verified independent samples. The provider exposes neither a seed nor backend routing guarantees; caching, updates, and shared infrastructure may induce dependence. V4 and V5 therefore report only their complete finite schedules. Their shared thresholds are fixed engineering choices, not estimated operating characteristics.

Each decision remains conditioned on its particular private registry.

The visible public test is intentionally useful and therefore can be overfit. The terminal private suite limits direct score adaptation but cannot guarantee semantic generalization beyond its 64 cases. Independent task-contract review, additional hidden registries, and adversarial cases would strengthen the record. The analyzer repairs are fully disclosed and outcome/count cross-checked, but remain post-registration changes to the analysis path. V5 repeated the same role-field defect already found in V4, exposing a regression-testing failure even though the narrow adapters and crosschecks preserve the reported decisions. Crosschecks do not rerun the scorers or independently reconstruct every registered file binding.

No human participants were studied. We do not measure understanding, time to reproduce a paper, or downstream scientific correctness. Evidence-carrying admission can reduce overconfident reuse, but a narrow pass can also create false assurance if its paper, task, provider, and scorer boundaries are omitted. Generated code remains a supply-chain risk; bounded workspaces and private scoring reduce but do not eliminate it.

Conclusion

EffectSlice treats a paper-derived core as a candidate substitution claim. It binds source-addressed provenance, visible public development, terminal private scoring, and a control-calibrated finite schedule into a local decision. Across 180 V4 conversations, controls pass, SNAP's registered full artifact misses sufficiency, and Toolformer prefix 01's slice misses despite sufficient F. A later schedule admits non-planted prefix 04 at 0/18/17 B/F/S successes. Across both schedules, 234 conversations yield, descriptively, one admission among three sequentially evaluated non-planted candidates. The contribution is bounded evidence for an auditable local admit/reject decision, not universal compression.

References

Dwork, C.; Feldman, V.; Hardt, M.; Pitassi, T.; Reingold, O.; and Roth, A. 2015. The Reusable Holdout: Preserving Validity in Adaptive Data Analysis. *Science*, 349(6248).

Gao, Y.; Li, Z.; Yuan, Y.; Ji, Z.; Ma, P.; and Wang, S. 2026. SkillReducer: Optimizing LLM Agent Skills for Token Efficiency. arXiv:2603.29919.

Groce, A.; Alipour, M. A.; Zhang, C.; Chen, Y.; and Regehr, J. 2016. Cause Reduction: Delta Debugging, Even without Bugs. *Software Testing, Verification and Reliability*, 26(2).

Jiang, H.; Wu, Q.; Lin, C.-Y.; Yang, Y.; and Qiu, L. 2023. LLMingua: Compressing Prompts for Accelerated Inference of Large Language Models. arXiv:2310.05736.

Liu, D.; Li, Z.; Du, H.; Wu, X.; Gui, S.; Kuang, Y.; and Sun, L. 2026. Graph-of-Skills: Dependency-Aware Structural Retrieval for Massive Agent Skills. arXiv:2604.05333.

Meng, X.; Wang, S.; and Fang, Y. 2026. SkillRAE: Agent Skill-Based Context Compilation for Retrieval-Augmented Execution. arXiv:2605.10114.

Miao, J.; Davis, J. R.; Zhang, Y.; Pritchard, J. K.; and Zou, J. 2025. Paper2Agent: Reimagining Research Papers as Interactive and Reliable AI Agents. arXiv:2509.06917.

Nakkiran, P.; and Blasiok, J. 2018. The Generic Holdout: Preventing False-Discoveries in Adaptive Data Science. arXiv:1809.05596.

Pan, Q.; Yang, Y.; Li, J.; Zhou, J.; Chen, K.; Li, X.; Chen, Q.; and He, L. 2026. Anything2Skill: Compiling External Knowledge into Reusable Skills for Agents. arXiv:2606.09316.

Pan, Z.; Wu, Q.; Jiang, H.; Xia, M.; Luo, X.; Zhang, J.; Lin, Q.; Ruhle, V.; Yang, Y.; Lin, C.-Y.; Zhao, H. V.; Qiu, L.; and Zhang, D. 2024. LLMingua-2: Data Distillation for Efficient and Faithful Task-Agnostic Prompt Compression. In *Findings of the Association for Computational Linguistics: ACL 2024*.

Regehr, J.; Chen, Y.; Cuoq, P.; Eide, E.; Ellison, C.; and Yang, X. 2012. Test-Case Reduction for C Compiler Bugs. In *Proceedings of the 33rd ACM SIGPLAN Conference on Programming Language Design and Implementation*, 335–346.

Schick, T.; Dwivedi-Yu, J.; Dessi, R.; Raileanu, R.; Lomeli, M.; Hambro, E.; Zettlemoyer, L.; Cancedda, N.; and Scialom, T. 2023. Toolformer: Language Models Can Teach Themselves to Use Tools. arXiv:2302.04761.

Shaposhnikov, M.; Fortuin, N.; Stipcich, S.; Gorinova, M. I.; Heineike, A.; and Willoughby, R. 2026. A Framework for Evaluating Agentic Skills at Scale. arXiv:2606.17819.

Shen, S.; Cheng, W.; Ma, M.; Turcan, A.; Zhang, M. J.; and Ma, J. 2026. SKILLFOUNDRY: Building Self-Evolving Agent Skill Libraries from Heterogeneous Scientific Resources. arXiv:2604.03964.

Starace, G.; Jaffe, O.; Sherburn, D.; Aung, J.; Chan, J. S.; Maksin, L.; Dias, R.; Mays, E.; Kinsella, B.; Thompson, W.; Heidecke, J.; Glaese, A.; and Patwardhan, T. 2025. PaperBench: Evaluating AI's Ability to Replicate AI Research. arXiv:2504.01848.

Xu, X.; and Wu, W. 2026. Compression, Structure, and Executor Capability: A Controlled Real-Cost Decomposition of Language-Model Agent Skill Optimisation. arXiv:2607.03048.

Yuan, B.; Su, Y.; Song, R.; Yang, S.; and Qin, J. 2026. ClawTrace: Cost-Aware Tracing for LLM Agent Skill Distillation. arXiv:2604.23853.

Zeller, A.; and Hildebrandt, R. 2002. Simplifying and Isolating Failure-Inducing Input. *IEEE Transactions on Software Engineering*, 28(2): 183–200.

Zhang, K.; Zemke, N. R.; Armand, E. J.; and Ren, B. 2024. A Fast, Scalable and Versatile Tool for Analysis of Single-Cell Omics Data. *Nature Methods*, 21: 217–227.

Zhao, X.; Sang, Z.; Li, Y.; Shi, Q.; Zhao, W.; Wang, S.; Zhang, D.; Han, X.; Liu, Z.; and Sun, M. 2025. AutoReproduce: Automatic AI Experiment Reproduction with Paper Lineage. arXiv:2505.20662.